

Build' a Tree®

Building teddy bears are no longer in. Instead, new green initiatives have made people more aware of green awareness. Therefore, you have started a new company - Build a Tree®. You and everyone who means something know that digital products written in Java are as cool as having a physical product. Thus, it is obvious that you should develop an awesome piece of Java code in VS Code.

Your thoughts

you haven't had that many ...

- 1) *"Maybe I should start with a class representing a Tree, where I can use the method `display()` to show current information about a tree, and a method `grow()` so that it can grow"*
- 2) *"I should therefore work with at least a height and an age, so that it can actually grow and get older! Additionally, I could parameterise `growth_rate` and a `max_age`, so that the tree can die. But I should also remember to keep an eye on whether or not a tree is alive or dead! And remember to print this also."*
- 3) *"Last, I could ensure that the tree flowers every 5th year, but not when it is every 20th year, unless of course we enter a new century!"*

You have drawn the following sketch to help you:

```
> Your tree is currently 0 years old, //display()
  It has reached the height of 1.0cm.
> And your tree just grew a year older! //grow()
> And your tree just grew a year older!
> And your tree just grew a year older!
> And your tree just grew a year older!
> And your tree just grew a year older!
> Your tree is currently 5 years old, //display()
```

```
It has reached the height of 32.0cm,  
and it is currently blooming.  
> And your tree just grew a year older! //grow()  
> And your tree just grew a year older!  
> And your tree just grew a year older!  
> And your tree just grew a year older!  
> And your tree just grew a year older!  
> And your tree just grew a year older!  
> The tree has died //grow()  
> The tree is dead, and reached the age 11 with a height of  
2048.0cm //display()  
> The tree is already dead. //trying to grow() again
```

(">" here implies a new line, but the formatting is up to you)

*Your hints as a non-experienced
programmer*

(You have none)